

Aerospace Academy Version

## Unit 2

### Technical Sketching for Aerospace Design

#### Unit Overview

Students develop visual communication skills by creating concept sketches, isometric sketches, multiview drawings, line convention examples, annotations, and dimensioned aerospace component drawing packages.

#### Essential Question

How do engineers communicate design ideas clearly before and after a part is modeled?

Name: \_\_\_\_\_ Period: \_\_\_\_\_ Date: \_\_\_\_\_

### Learning Targets

1. Explain the major concepts of technical sketching for aerospace design.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

### Unit Deliverables

- Sketching practice pages
- Isometric aerospace component sketch
- Multiview drawing practice
- Line conventions reference
- Dimensioning practice
- Annotated drawing
- Final aerospace component drawing package

## Vocabulary

| Term                           | Definition |
|--------------------------------|------------|
| <b>Concept Sketch</b>          |            |
| <b>Isometric Sketch</b>        |            |
| <b>Orthographic Projection</b> |            |
| <b>Multiview Drawing</b>       |            |
| <b>Front View</b>              |            |
| <b>Top View</b>                |            |
| <b>Right-Side View</b>         |            |
| <b>Object Line</b>             |            |
| <b>Hidden Line</b>             |            |
| <b>Centerline</b>              |            |
| <b>Dimension Line</b>          |            |
| <b>Extension Line</b>          |            |
| <b>Annotation</b>              |            |
| <b>Scale</b>                   |            |
| <b>Drawing Package</b>         |            |

**Unit Project: Aerospace Component Drawing Package**

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

## Day 1 — Why Engineers Sketch

**Objective:** I can explain how sketching helps engineers communicate design ideas.

**Bellwork:** Why might engineers sketch before using CAD?

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**Core Task:** Compare a rough sketch, a detailed sketch, and a technical drawing. Identify what each communicates.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What can a sketch communicate that words alone cannot?

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**Day 2 — Concept Sketching**

**Objective:** I can create quick concept sketches with labels.

**Bellwork:** What makes a sketch useful even if it is not perfect?

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**Core Task:** Sketch three possible aerospace components and label major features.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What label made your sketch easier to understand?

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### Day 3 — Sketching with Annotations

**Objective:** I can add annotations that explain design intent.

**Bellwork:** What information should annotations explain?

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**Core Task:** Upgrade a concept sketch with design intent notes, materials, and function descriptions.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What is one strong annotation you added?

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### Day 4 — Isometric Sketching Basics

**Objective:** I can sketch objects using isometric axes.

**Bellwork:** Why are 3D sketches useful for engineering communication?

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**Core Task:** Practice sketching simple blocks and aerospace-style brackets on isometric grid paper.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What is difficult about isometric sketching?

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**Day 5 — Isometric Aerospace Component**

**Objective:** I can create an isometric sketch of an aerospace component.

**Bellwork:** How can shape communicate function?

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**Core Task:** Create an isometric sketch of a wing rib, sensor mount, fin bracket, or payload container.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What feature makes your component aerospace-related?

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**Day 6 — Multiview Drawing Basics**

**Objective:** I can explain front, top, and side views.

**Bellwork:** Why might a 3D sketch not be enough to manufacture a part?

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**Core Task:** Match objects to front/top/right-side views and create basic multiview sketches.

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**Evidence of Work / Notes:**

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**Exit Ticket:** Which view usually shows the most detail?

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**Day 7 — Multiview Practice**

**Objective:** I can create aligned multiview drawings.

**Bellwork:** Why must views be aligned?

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**Core Task:** Create a three-view drawing of a simple aerospace component.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What view was hardest to draw?

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**Day 8 — Line Conventions**

**Objective:** I can identify and use engineering line types.

**Bellwork:** Why do engineers use different line types?

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**Core Task:** Create a line convention reference with object, hidden, center, dimension, and extension lines.

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**Evidence of Work / Notes:**

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**Exit Ticket:** Which line type communicates information that cannot be seen from the outside?

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**Day 9 — Dimensioning Fundamentals**

**Objective:** I can add dimensions clearly and correctly.

**Bellwork:** Why do manufacturers need dimensions?

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**Core Task:** Dimension a simple multiview drawing using extension and dimension lines.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What makes a dimension unclear?

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**Day 10 — Annotations and Design Notes**

**Objective:** I can add notes that clarify function and design intent.

**Bellwork:** When should a drawing include notes instead of only dimensions?

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**Core Task:** Add notes about material, function, payload fit, and manufacturing concerns.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What note would help a manufacturer?

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**Day 11 — Drawing Package Planning**

**Objective:** I can plan a complete aerospace drawing package.

**Bellwork:** What should be included in a drawing package?

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**Core Task:** Select a component and plan the final sketch/drawing package requirements.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What drawing will be most important in your package?

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**Day 12 — Drawing Package Work Day 1**

**Objective:** I can create accurate sketches for my drawing package.

**Bellwork:** What does quality work look like in a technical drawing?

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**Core Task:** Work on concept sketch and isometric sketch for final component.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What part of your drawing needs revision?

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**Day 13 — Drawing Package Work Day 2**

**Objective:** I can create a multiview drawing with dimensions.

**Bellwork:** How can you check if your views match?

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**Core Task:** Work on multiview drawing and dimensions. Peer-check alignment and clarity.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What feedback did you receive?

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**Day 14 — Drawing Package Revision**

**Objective:** I can revise a technical drawing using feedback.

**Bellwork:** Why is revision part of technical communication?

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**Core Task:** Revise sketches, annotations, line conventions, and dimensions.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What did you improve?

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**Day 15 — Drawing Package Submission**

**Objective:** I can submit and reflect on a complete aerospace drawing package.

**Bellwork:** What makes a drawing package professional?

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**Core Task:** Submit final drawing package and complete reflection.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What skill improved most during this unit?

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## Unit 2 Submission Checklist

| Item                                      | Complete?                |
|-------------------------------------------|--------------------------|
| Sketching practice pages                  | <input type="checkbox"/> |
| Isometric aerospace component sketch      | <input type="checkbox"/> |
| Multiview drawing practice                | <input type="checkbox"/> |
| Line conventions reference                | <input type="checkbox"/> |
| Dimensioning practice                     | <input type="checkbox"/> |
| Annotated drawing                         | <input type="checkbox"/> |
| Final aerospace component drawing package | <input type="checkbox"/> |

## Unit 2 Project Rubric

| Category                                       | Points     |
|------------------------------------------------|------------|
| Technical accuracy and completeness            | 20         |
| Engineering process and documentation          | 20         |
| Aerospace connection and design intent         | 15         |
| Quality of sketches, models, data, or drawings | 20         |
| Use of evidence, testing, or decision making   | 15         |
| Reflection and professionalism                 | 10         |
| <b>Total</b>                                   | <b>100</b> |

## Unit 2 Student Work Templates

## Concept Sketch Practice

**Purpose:** Concept development

**Directions:** Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

**Concept title:** \_\_\_\_\_

**Design goal:** \_\_\_\_\_

**Key features:** \_\_\_\_\_

### Aerospace Component Concept Sketch

**Strongest** \_\_\_\_\_

**feature:** **Biggest concern:** \_\_\_\_\_



## Isometric Practice Page 1

**Purpose:** Isometric sketching

**Directions:** Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Purpose:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Notes / annotations:** \_\_\_\_\_

## Isometric Practice Page 2

**Purpose:** Isometric sketching

**Directions:** Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Purpose:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Notes / annotations:** \_\_\_\_\_

## Multiview Practice Page

**Purpose:** Orthographic / multiview sketching

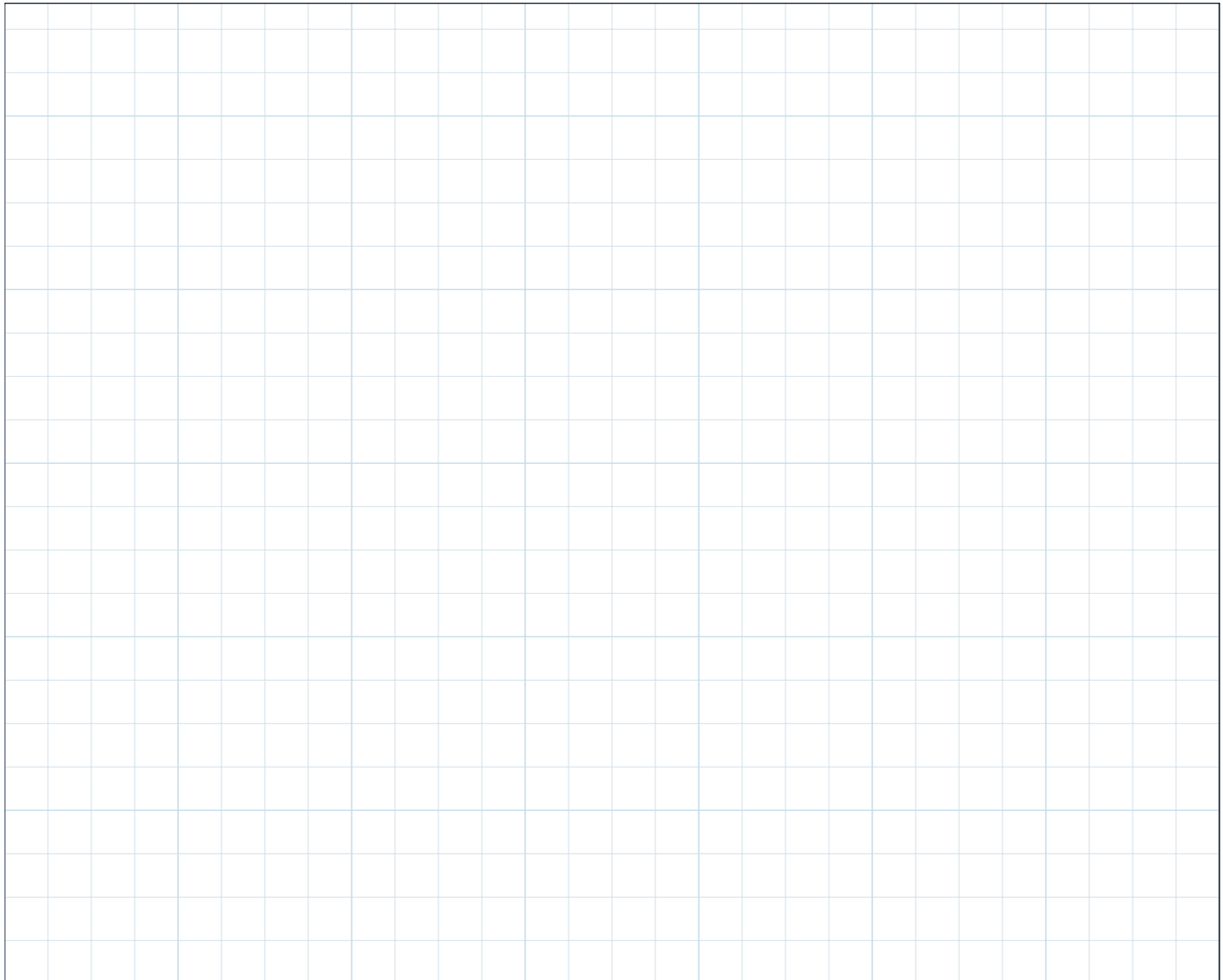
**Directions:** Use the orthographic grid to align the front, top, and right-side views. Show only the views requested. Use projection alignment and line conventions.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Main view:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Dimension / line convention notes:** \_\_\_\_\_



## Dimensioning Practice Page

**Purpose:** Orthographic / multiview sketching

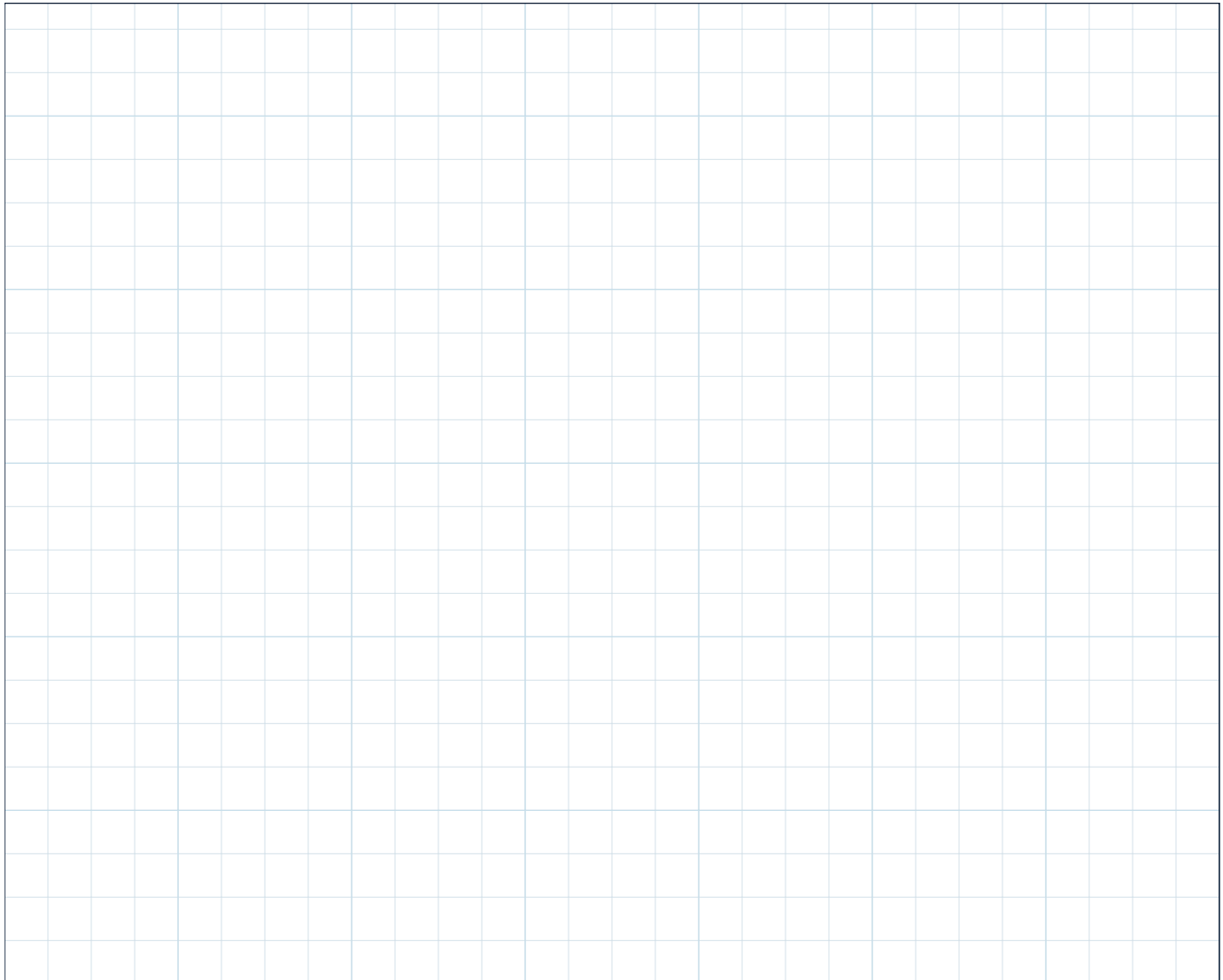
**Directions:** Use the orthographic grid to align the front, top, and right-side views. Show only the views requested. Use projection alignment and line conventions.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Main view:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Dimension / line convention notes:** \_\_\_\_\_

## Final Drawing Package Page

**Purpose:** Orthographic / multiview sketching

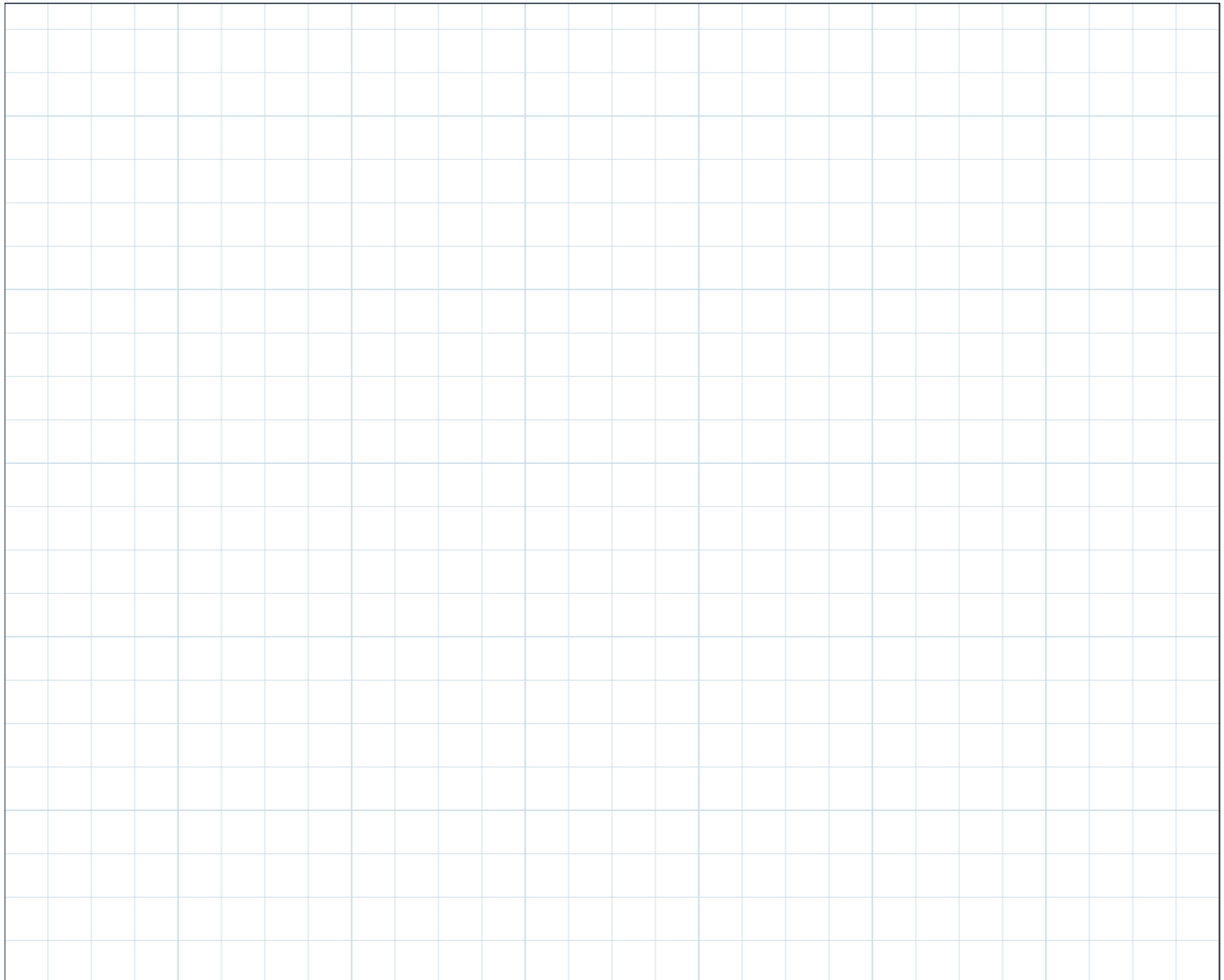
**Directions:** Use the orthographic grid to align the front, top, and right-side views. Show only the views requested. Use projection alignment and line conventions.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Main view:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Dimension / line convention notes:** \_\_\_\_\_

## Line Convention Reference

**Purpose:** Technical communication

**Directions:** Draw and label each engineering line type. Include a short note about when it is used.

|                   |       |
|-------------------|-------|
| Object line       | _____ |
| Hidden line       | _____ |
| Centerline        | _____ |
| Dimension line    | _____ |
| Extension line    | _____ |
| Construction line | _____ |

